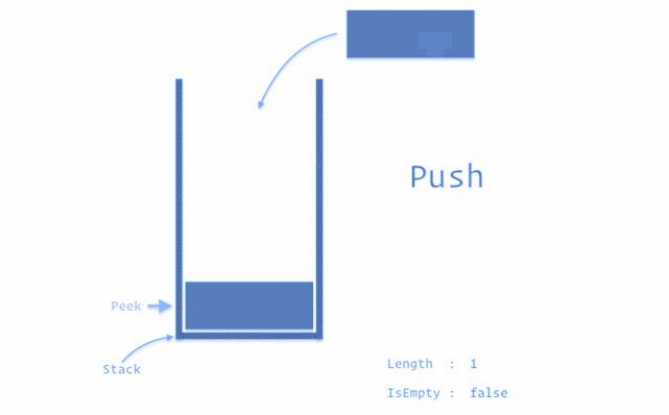
Stacks, Queues and Monotonicity

Pre-requisites

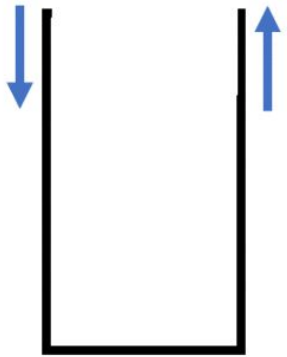
- Linear data structures array/list
- Linked List

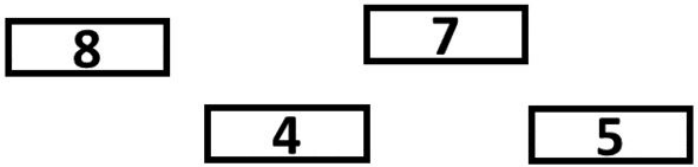
Stacks



Introduction To Stack

Stack data structure is a linear data structure that accompanies a principle known as **LIFO** (Last In First Out) or FILO (First In Last Out).





Real Life Example

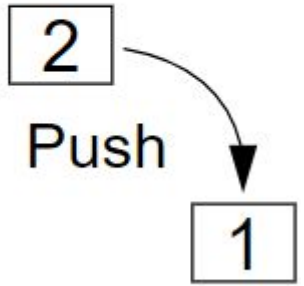
Stack of plates

Stack of books

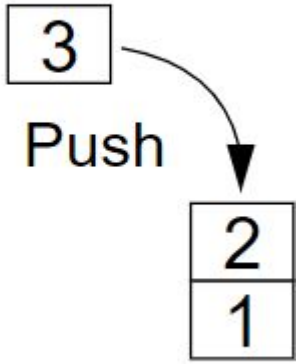
Stack Operations

Push operation

- Add an element to the top of the stack



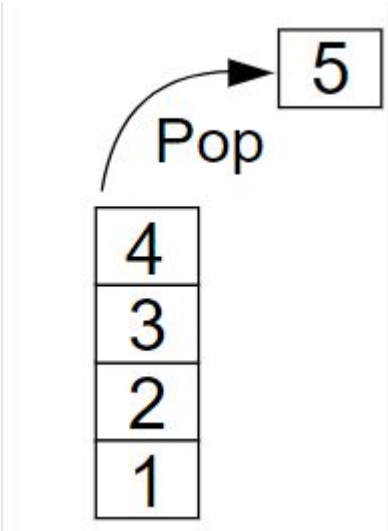
Stack Operations



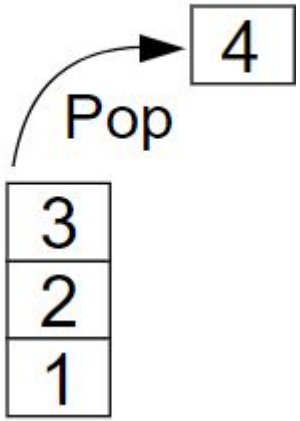
Stack Operations

Pop operation

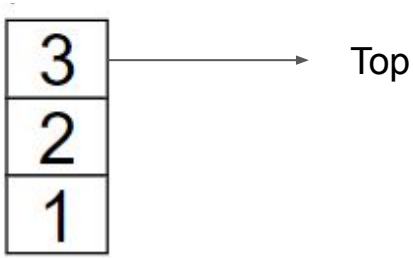
- Remove element from the top of the stack



Stack Operations



Stack Operations



Peek operation

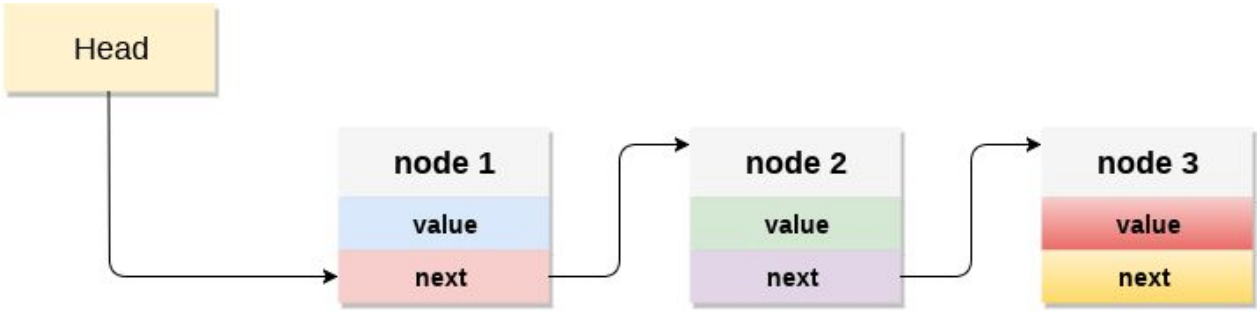
- returning the top element of a stack.

Is empty()

- Check if the list is empty or not.
- If it's empty return True else False.

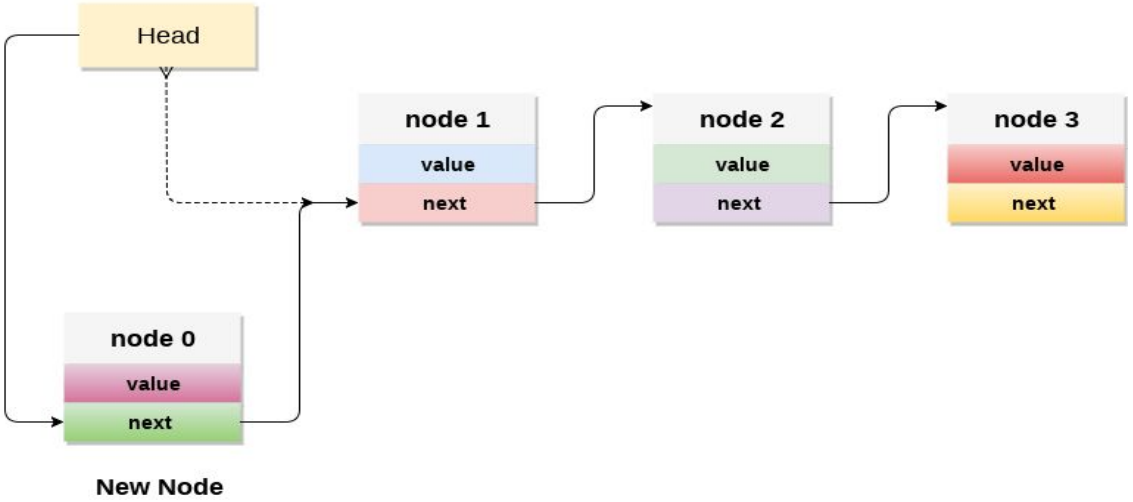
How can we implement Stack?

Implementing stack using linked list



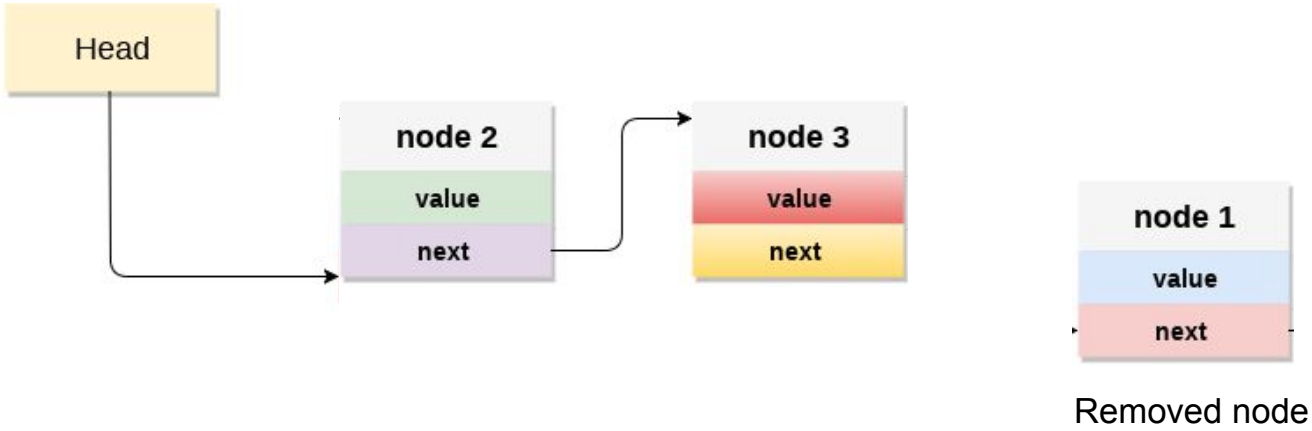
Push operation

- Initialise a node
- Update the value of that node by data i.e. node.value = data
- Now link this node to the top of the linked list i.e. node.next = head
- And update top pointer to the current node i.e. head = node



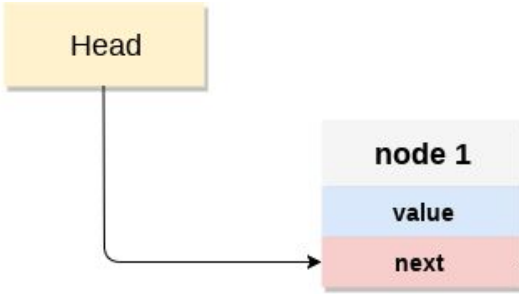
POP operation

- First Check whether there is any node present in the linked list or not, if not then return
- Otherwise make pointer let say temp point to the top node and move forward the top node by 1 step
- Now free this temp node



Top operation

- Check if there is any node present or not, if not then return.
- Otherwise return the value of top node of the linked list which is the node at **Head**



Implementation

```
class Node:
     def __init__(self, data):
          self.data = data
          self.next = None

class Stack:
     def __init__(self):
          self.head = None

     def isempty(self):
          if self.head == None:
               return True
          else:
               return False

     def peek(self):
          if self.isempty():
               return None
          else:
               return self.head.data
```

```
def push(self, data):
##G5C is great
     if self.head == None:
          self.head = Node(data)
     else:
          new_node = Node(data)
          new_node.next = self.head
          self.head = new_node

def pop(self):

     if self.isempty():
          return None
     else:
          popped_node = self.head
          self.head = self.head.next
          popped_node.next = None
          return popped_node.data
```

Time and space complexity

- Push
  - Time complexity - ___?
- Pop
  - Time complexity - ___?
- Peek
  - Time complexity - ___?
- isEmpty()
  - Time complexity - ___?

Time and space complexity

- Push
  - Time complexity - O(1)
- Pop
  - Time complexity - O(1)
- Peek
  - Time complexity - O(1)
- isEmpty()
  - Time complexity - O(1)

Applications of Stack

Practice

Problem

Solution

```python
class Solution:
    def isValid(self, s: str) -> bool:
        stack = []
        my_dict = {"(" : ")", "{" : "}", "[" : "]"}
        for i in range(len(s)):
            if s[i] in my_dict.keys():
                stack.append(s[i])
            else:
                if not stack:
                    return False
                a = stack.pop()
                if s[i] != my_dict[a]:
                    return False
        return stack == []
```

Practice

Problem

Solution

```python
class Solution:
    def removeStars(self, s: str) -> str:
        N = len(s)
        stack = []
        for i in range(N):
            if s[i] == "*":
                if stack:
                    stack.pop()
            else:
                stack.append(s[i])
        return ''.join(stack)
```

Reflection: Stack can help you simulate deletion of elements in the middle in O(1) time complexity

Pair Programming

Problem

Solution

```python
class Solution:
    def minOperations(self, logs: List[str]) -> int:
        stack = []
        for log in logs:
            if log == '../':
                if stack:
                    stack.pop()
            elif log == './':
                continue
            else:
                stack.append(log)
        return len(stack)
```

Reflection: stack can help you defer decision until some tasks are finished.

The bottom of the stack waits on the top of stack until they are processed

Common PitFalls

- Popping from empty list
  - This will throw index out of range error
- Null pointer exception if we are using linked list

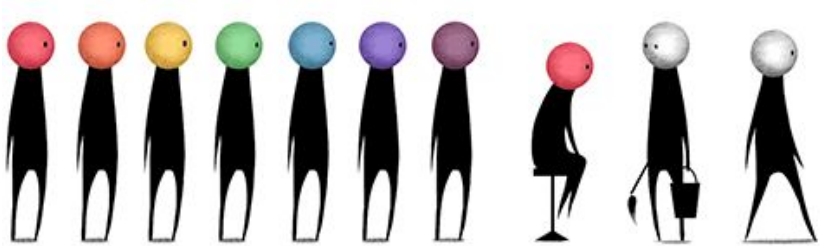
Runtime Error

```
IndexError: list index out of range
    if i == open_close[stack[-1]]:
Line 6 in isValid (Solution.py)
    ret = Solution().isValid(param_1)
Line 32 in _driver (Solution.py)
    _driver()
Line 43 in <module> (Solution.py)
```

Common PitFalls

- Stack overflow
    - May be not in python but In other programming language
    - Pushing to a full stack

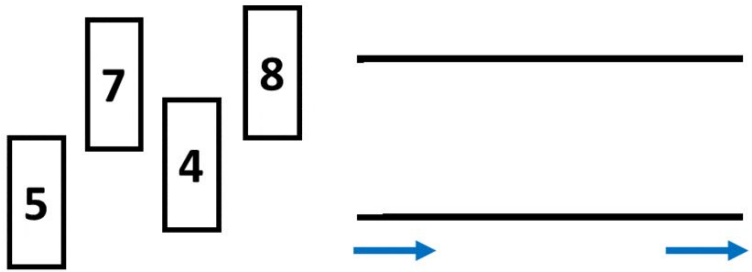
Queues



Introduction

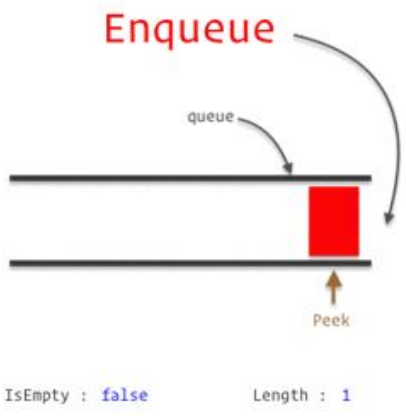
A collection whose elements are added at one end (the **rear**) and removed from the other end (the **front**)

- Uses **FIFO** data handling



Real Life Example

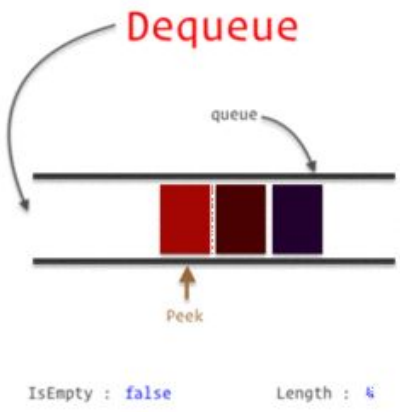
Queue Operations



Enqueue (Append)

- Add an element to the tail of a queue
- **F**irst **I**n

Queue Operations



Dequeue (Popleft)

- Remove an element from the head of the queue
- **F**irst **O**ut

Practice

Problem

Solution

```python
class RecentCounter:
    def __init__(self):
        self.queue = []

    def ping(self, t: int) -> int:
        self.queue.append(t)
        while (t - self.queue[0]) > 3000:
            self.queue.pop(0)
        return len(self.queue)
```

Implementing Queue

Using an array to implement a queue is significantly harder than using an array to implement a stack. **Why**?

What would the time complexity be?

Implementing Queue with List

```
def __init__(self):
   self.queue = []
   self.headIndex = 0

def append(self, value: int):
   self.queue.append(value)

def pop(self) -> int:
   if self.headIndex < len(self.queue):
        val = self.queue[self.headIndex]
        self.headIndex += 1
        return val
```

A More Optimal Implementation

```
def __init__(self, size: int):
    self.queue = [None] * size
    self.headIndex = 0
    self.tailIndex = 0
    self.size = size
    self.count = 0

def append(self, value: int) -> bool:
    if self.count == self.size:
        return False
    self.queue[self.tailIndex] = value
    self.tail = (self.tailIndex + 1) % self.size
    self.count += 1
    return True
```

Continued ...

```
def pop(self) -> int:
    if self.count == 0:
        None
    value = self.queue[self.headIndex]
    self.queue[self.headIndex] = None
    self.headIndex = (self.headIndex + 1) % self.size
    self.count -= 1
    return value
```

Implementing Queue

- Either linked list or list can be used with careful considerations
- In practice, prefer to use built-in or library implementations like `deque()`
- Internally, `deque()` is implemented as a linked list of nodes

```
.pop()
.append()
.popleft()
.appendleft()
```

Implementation (built-in)

```
from collections import deque
# Initializing a queue
queue = deque()
# Adding elements to a queue
queue.append('a')
queue.append('b')
# Removing elements from a queue
print(queue.popleft())
print(queue.popleft())
# Uncommenting queue.popleft()
# will raise an IndexError
# as queue is now empty
```

We can also use it the other way around by using;

- `.appendleft()`
- `.pop()`

Time and space complexity

- Append
  - Time complexity - ____?
- Popleft
  - Time complexity - ____?
- Peek
  - Time complexity - ____?
- isEmpty()
  - Time complexity - ____?

Time and space complexity

- Append
  - Time complexity - O(1)
- Popleft
  - Time complexity - O(1)
- Peek
  - Time complexity - O(1)
- isEmpty()
  - Time complexity - O(1)

Applications of Queue

Practice

Problem

Solution

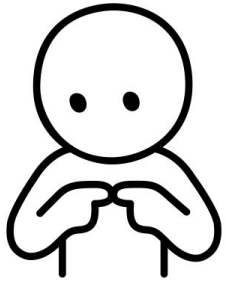
```python
from collections import deque
class DataStream:
    def __init__(self, value: int, k: int):
        self.value = value
        self.k = k
        self.deque = deque()
        self.count = 0

    def consec(self, num: int) -> bool:
        if len(self.deque) == self.k:
            if self.deque[0] == self.value:
                self.count -= 1
            self.deque.popleft()

        self.deque.append(num)
        if num == self.value:
            self.count += 1

        return self.count == self.k
```

Reflection: Queue helps solve problems that need access to the “first something”



Common Pitfalls

Africa To Silicon Valley

Africa To Silicon Valley

Not handling edge cases

- Popping from an empty queue
  - ```
    if queue:

        queue.popleft()
    ```

- Appending to a full queue
  - ```
    if len(queue) < capacity:

        queue.append(val)
    ```

Practice Questions

Stacks

- Valid Parentheses
- Simplify Path
- Evaluate Reverse Polish Notation
- Score of parenthesis
- Backspace String Compare

Queues

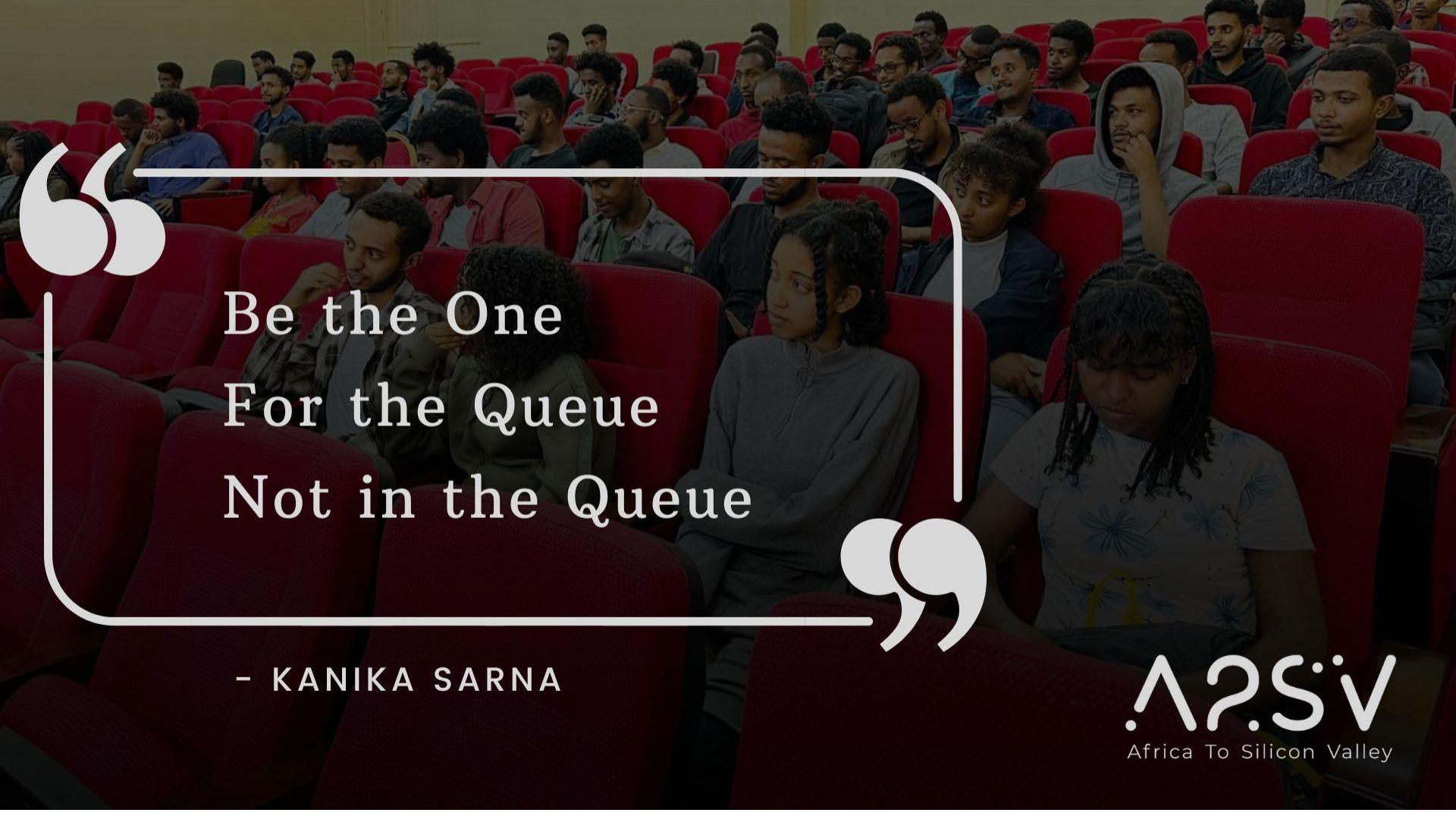
- Number of recent calls
- Find consecutive integers
- Design Circular Deque
- Implement Queue using Stack
- Shortest subarray with sum at least K

Monotonic

- Car Fleet
- Remove duplicates
- Sum of subarray minimum
- Remove k digits
- 132 Pattern

Resources

A comprehensive guide and template for monotonic stack based problems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